

# Kmos-10010

## SUMMARY

---

Purpose: This document defines the complete implementation roadmap for the Knowledge & Media Operating System (KMOS). Unlike the architecture specifications, this document organizes implementation into executable engineering work. It defines build order, milestones, deliverables, dependencies, acceptance criteria, and validation strategy.

## VERSION

---

1.0

## STATUS

---

Master Engineering Roadmap

## ENGINEERING PHILOSOPHY

---

Implementation SHALL proceed by proving architecture through working software. The roadmap SHALL prioritize correctness, observability, governance, maintainability, and institutional preservation. Optimization SHALL occur only after correctness has been demonstrated.

## MASTER MILESTONES

---

M0 — Engineering Foundation; M1 — Foundational Institutional Engines; M2 — Capability Execution Platform; M3 — Domain Services; M4 — Applications; M5 — Production Hardening; M6 — Reference Platform Certification. Each milestone SHALL conclude with a deployable platform.

## MILESTONE M0 — ENGINEERING FOUNDATION

---

Objective: Establish the engineering environment required for KMOS. Deliverables: Repository structure; Coding standards; CI/CD pipeline; Testing framework; Documentation framework; Container environment; Deployment automation; Developer tooling; Architecture validation scripts. Acceptance Criteria: Repository initializes successfully; CI executes automatically; Test framework operational; Documentation generation operational; Local development environment reproducible.

## MILESTONE M1 — FOUNDATIONAL INSTITUTIONAL ENGINES

---

Objective: Implement the seven foundational engines. M1.1 Knowledge Service (Reference: KMOS-0201) Deliverables: Knowledge Repository; Knowledge APIs; Version Management; Relationship Engine; Vocabulary Manager; Knowledge Events. Acceptance: Service passes all KMOS-0201 acceptance criteria. M1.2 Asset Registry (Reference: KMOS-0202) Deliverables: Asset Registry; Lineage Engine; Provenance; Evidence Packages; Metadata Management; Integrity Verification. Acceptance: Complete derivative tracking. M1.3 Event Service (Reference: KMOS-0203) Deliverables: Canonical Event Registry; Replay Engine; Subscriptions; Routing; Correlation; Dead Letter Management. Acceptance: Replay validated. M1.4 Identity Service (Reference: KMOS-0206) Deliverables: Identity Registry; Authentication; Authorization; Organizations; Service Accounts; AI Worker Identities. Acceptance: Every platform component possesses canonical identity. M1.5 Governance Service (Reference: KMOS-0207) Deliverables: Policy Registry; Approval Engine; Certification; Audit; Trust Assessment; Risk Management. Acceptance: Evidence-based governance operational.

## MILESTONE M2 — CAPABILITY EXECUTION PLATFORM

---

Objective: Enable execution of business capabilities. M2.1 Capability Registry (Reference: KMOS-0205)  
Deliverables: Capability Discovery; Manifest Validation; Certification; Dependency Graph; Versioning. Acceptance: Capabilities dynamically discoverable. M2.2 Capability Runtime Deliverables: Execution Runtime; Isolation; Scaling; Health; Configuration. Acceptance: Capabilities execute independently. M2.3 Workflow Service (Reference: KMOS-0204) Deliverables: Workflow Registry; Execution Engine; Scheduling; Human Tasks; Compensation. Acceptance: Long-running workflows operational. M2.4 Configuration Service Deliverables: Configuration Registry; Secrets Integration; Versioning; Profiles; Overrides. Acceptance: External configuration operational. M2.5 Search Service Deliverables: Semantic Search; Asset Search; Knowledge Search; Capability Search; Governance Search. Acceptance: Cross-service discovery operational.

---

## **MILESTONE M3 — DOMAIN SERVICES**

Reference services include: Media Service; Language Service; Publishing Service; Preservation Service; AI Collaboration Service; Connector Framework. Each SHALL reuse foundational engines.

---

## **MILESTONE M4 — APPLICATIONS**

Applications SHALL remain thin. Examples: Studio; Research Portal; Archive Explorer; Mobile Applications; Administration. Applications SHALL contain minimal business logic.

---

## **MILESTONE M5 — PRODUCTION HARDENING**

Deliverables: Performance Optimization; Security Review; Scalability Validation; Disaster Recovery; Monitoring; Operational Documentation; Migration Testing; Certification.

---

## **MILESTONE M6 — REFERENCE PLATFORM CERTIFICATION**

Reference applications SHALL validate: Media Production; Knowledge Preservation; Publishing; Learning; Research. The architecture SHALL prove itself through operational systems.

---

## **ENGINEERING WORK PACKAGE TEMPLATE**

Every implementation work package SHALL define: Objective; Specifications; Dependencies; Deliverables; Automated Tests; Acceptance Criteria; Documentation; Operational Evidence; Governance Review. This template SHALL be reused across all implementation work.

---

## **DEPENDENCY MATRIX**

Knowledge; Assets; Events; Identity; Governance -> Capability Registry -> Capability Runtime -> Workflow -> Configuration -> Search -> Domain Services -> Applications. No milestone SHALL violate dependency ordering without documented justification.

---

## **DEFINITION OF DONE**

A work package is complete only when: Implementation complete; Tests passing; Documentation complete; Events validated; Business contracts verified; Observability operational; Governance approved; Deployment verified; Production readiness demonstrated.

---

## **AI CODING AGENT OPERATING MODEL**

An autonomous implementation agent SHALL: Complete one work package at a time; Run automated verification; Commit working increments; Generate documentation continuously; Never leave failing tests unresolved; Stop only when architectural ambiguity exists.

## **PROGRESS TRACKING**

---

Every milestone SHALL record: Planned Work; Completed Work; Remaining Work; Known Risks; Technical Debt; Architecture Deviations; Governance Decisions; Implementation Evidence. Progress SHALL be evidence-based.

## **SUCCESS CRITERIA**

---

The roadmap is complete when: All foundational engines are operational; Capability execution is operational; Workflow execution is operational; Domain services are operational; Reference applications validate the architecture; The platform satisfies all constitutional principles. Implementation SHALL conclude with a deployable KMOS platform rather than a collection of disconnected services.